Eduardo Jose Duarte Peña

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eduardo-duarte-penna • Mechatronics Engineer | Data Analytics & Machine Learning

Education

Universidad Santo Tomás — **Colombia**
B.Sc. in Mechatronics Engineering — *2019–2025*

Coursera - Google
Data Analytics Specialization — *2023*

Professional Experience

Space-Eyes
Machine Learning Engineer — *Oct 2025 – Present*

- Designed and implemented an **end-to-end RF data acquisition and processing pipeline** in Python, capturing wideband IQ signals and generating spectrograms for robust signal analysis.
- Developed a **real-time drone detection and triangulation system** using Raspberry Pi, integrating machine learning models with specialized RF hardware (SDRs, GPSDO, and long-range directional WiFi antennas) for high-stability communication and accurate localization.
- Defined and documented the **mathematical foundations for RF-based triangulation**, ensuring reproducibility and precision across diverse operational scenarios.
- Implemented a **resilient MQTT-based distributed communication architecture**, achieving stable telemetry and RF data transmission over distances up to 5 km under challenging RF conditions.
- Built a **scalable backend using FastAPI** to process and expose hardware-acquired RF data through structured APIs, enabling real-time interaction and monitoring.
- Leveraged **Microsoft Azure** for training and optimizing machine learning models, improving detection performance and enabling scalable experimentation workflows.
- Developed a **React-based frontend** for real-time visualization of RF data, system status, and geospatial drone tracking.
- Managed **version control and collaborative workflows using GitHub**, ensuring traceability and structured development.
- Led the transition from **MVP to Tier-1 system architecture**, defining system tiers, identifying bottlenecks, and implementing improvements in scalability, reliability, and modularity.

ELVA AI
Machine Learning Engineer — *Aug 2025 – Sept 2025*

- Trained a model to classify Wi-Fi, LTE, and DVB-T signals from IQ samples, optimized for resource-constrained devices.
- Automated city-level reports including confusion matrices and interactive visualizations.
- Implemented authentication in AKS with Azure AD, improving cloud deployment security.
- Reduced storage costs by 20% through disk optimization and retention policies in Azure.

TAYRONAMA SAS
Automation Developer & Data Analytics — *Mar 2025 – Jul 2025*

- Developed a Python application for pig traceability, integrating data acquisition through serial communication.
- Processed and analyzed data using Pandas, generating automated reports and Power BI dashboards.
- Implemented automated report delivery via WhatsApp API, improving client communication efficiency.

Merck Sharp & Dohme (MSD)
Innovation and Technology Intern - Data Analytics — *Jul 2024 – Feb 2025*

- Saved more than 30 hours per month by optimizing training processes through data analysis.
- Developed a KPI monitoring system in Power BI to ensure performance quality.
- Implemented solutions using Power Apps and Power Automate, improving operational efficiency.
- Processed large datasets using Python, SQL, and machine learning libraries.

Universidad Santo Tomás
Data Scientist / Junior Researcher — *Jun 2023 – Nov 2023*

- Built machine learning models to predict public transportation routes.
- Developed a Power BI dashboard with urban mobility insights for Bucaramanga.
- Scientific publication: *Advances in solutions on public transportation problems with AI*.

Courses and Certifications

2024: Process Improvement, Harvard ManageMentor

2023: Data Science & AI, MinTIC

2023: Machine Learning Fundamentals, DataCamp

2023: Implementing AI Solutions in Business, DataCamp

2023: Databases and SQL for Data Science, Coursera

2023: Deep Learning in Python, DataCamp

2022: Python for Data Science, Coursera

Technical Skills

Programming: Python, SQL, React, FastAPI

Machine Learning: Machine Learning, Deep Learning, Signal Classification, Model Optimization, PyTorch

RF & Signal Processing: IQ Signal Processing, Spectrogram Generation, RF Spectrum Analysis, SDR (Software Defined Radio), bladeRF

Data & Analytics: Data Analysis, Pandas, Data Visualization, Power BI

Systems & Infrastructure: MQTT Distributed Systems, Data Pipelines, Edge Computing, Cloud Deployment (Azure, AKS)

Tools: Power BI, Power Apps, Power Automate, Git

Methodologies: CRISP-DM, Process Automation

Awards

2023: 1st Place, CienTIoN – Research and Innovation Category

2023: 3rd Place, Huawei ICT Competition - Cloud Track

Languages

Spanish: Native

English: B2

References

Sebastián Murcia – General Manager, Tayronama SAS – +57 321 6680277

Iván Lis – Safety Data Manager, MSD – +57 310 7735259

Carlos Pimentel – Research Professor, Universidad Santo Tomás – +57 316 2373049